



T16 Dealer Repair Training



Module Principle

T16 Cable Connection

T16 Damage Assessment

T16 Disassembly

FAQ



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Aircraft Diagram



Aircraft Diagram



The No. 1, 3, and 6 motor are CCW motors, and the No. 2, 4, and 5 motors are CW motors. The spray lance is under the No. 2, 3, 5, and 6 motors' arm, and there are two sprinklers. The aerial-electronics board, spraying board, ESC board, and radar have a protection rating of IP67.

Modules

①



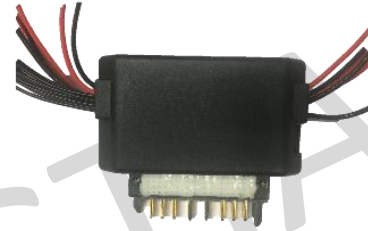
②



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⑫





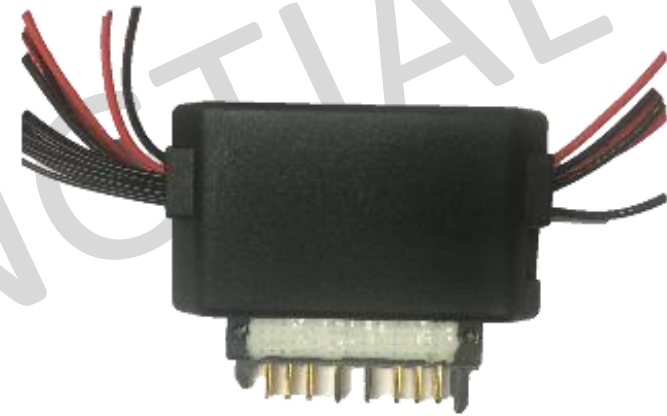
① The **Aerial-Electronics Board** is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. It manages the aircraft's resources.



② **Rotating Radar**: The Rotating Radar is the obstacle avoidance module, and it serves as the sensor when the aircraft is flying. It detects the front, lower, and rear obstacles, and it sends information to the aerial-electronics board.



③ **LED Board**: The LED board indicates the aircraft's current working condition. Users can know the aircraft's current working condition by the LED's color.



④ The **Power Distribution Board** sends power to the aircraft's modules and sends battery information to the aerial-electronics board.



⑤ **Spray Lance**



⑥ **RTK Module**: The RTK Module provides the aircraft's location, direction, and speed information to the flight controller system. It ensures the aircraft's stability and operation accuracy.



⑦ **Motor:** The motors work with propellers to blow wind downwards and make the aircraft fly upwards. It also helps with spraying. The six motors are interchangeable.



⑧ **ESC:** The ESC board converts the throttle information to the motor's rotation speed information, controls the motor rotation, and sends the motor information to the aerial-electronics board. The six ESCs are interchangeable.



⑨ **Delivery Pump:** The Delivery Pump controls the aircraft's spraying efficiency. Motors and ESC modules are included.



⑩ **Flow Meter:** The Flow Meter is essential for high-precision spraying. It detects the pump's spraying speed and reports the information to the aerial-electronics board. Therefore, the pump's spraying accuracy can be controlled.

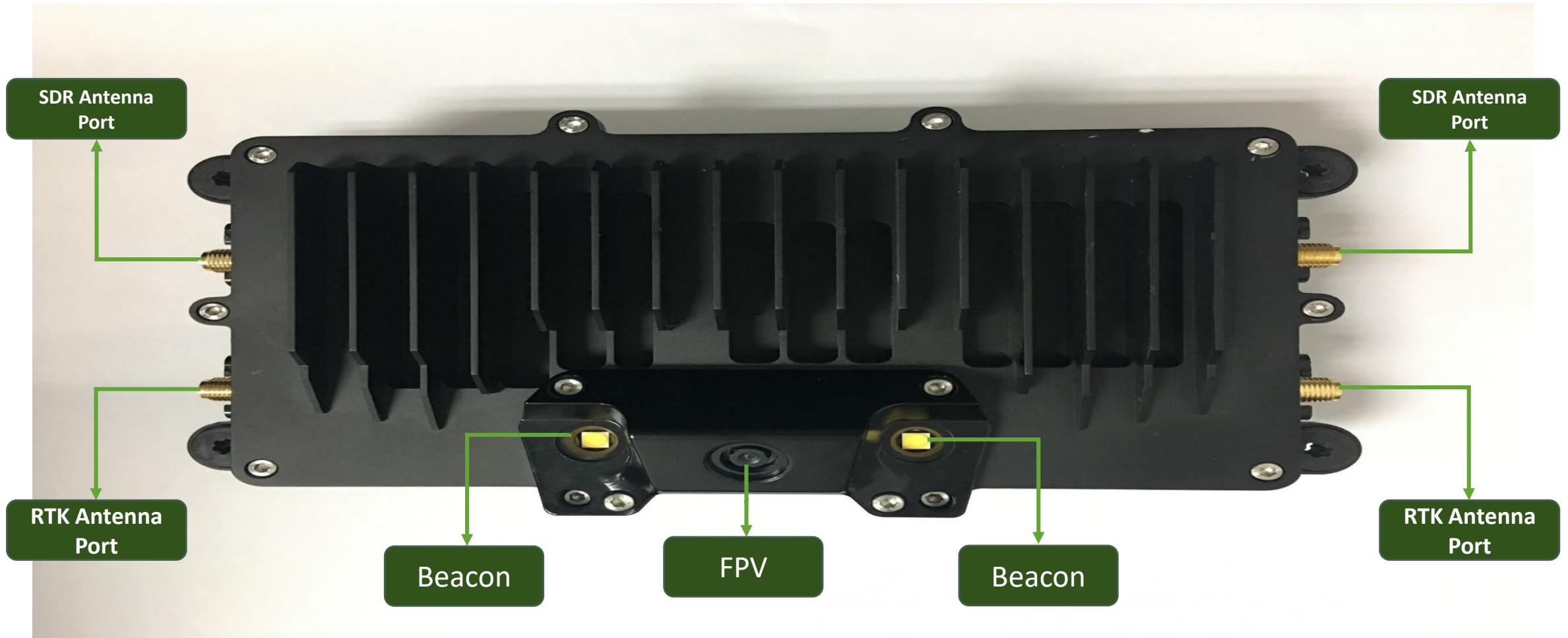


⑪ **Spraying Board:** The Spraying Board is the control system of the aircraft's core spraying function. It controls the delivery pump and spraying volume.



⑫ **4-channel Intelligent Battery Charger and Battery:** The battery is the energy source of flying and spraying. It stores energy.

Aerial-Electronics Board - Front



The Aerial-Electronics Board is the core module controlling the flight controller. It integrates the flight algorithm, image transmission, and flight controlling functions into one module. It manages the aircraft's resources. (Flight controller module, 1860, SDR module, barometer, FPV, and intelligent memory module.)

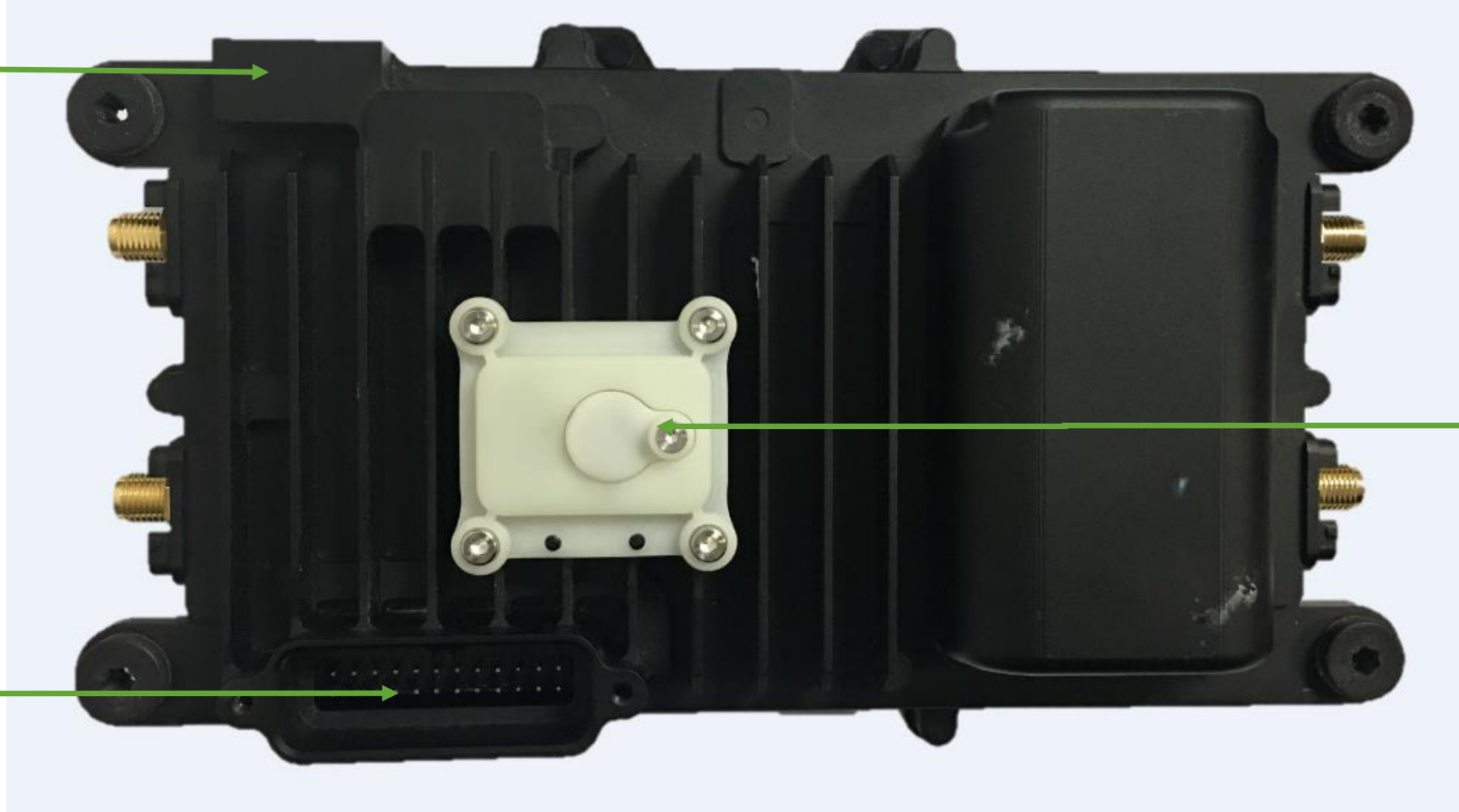
Note: Do not damage the flat cable when assembling or disassembling the front vision module.

Aerial-Electronics Board - Back

1. Update Port

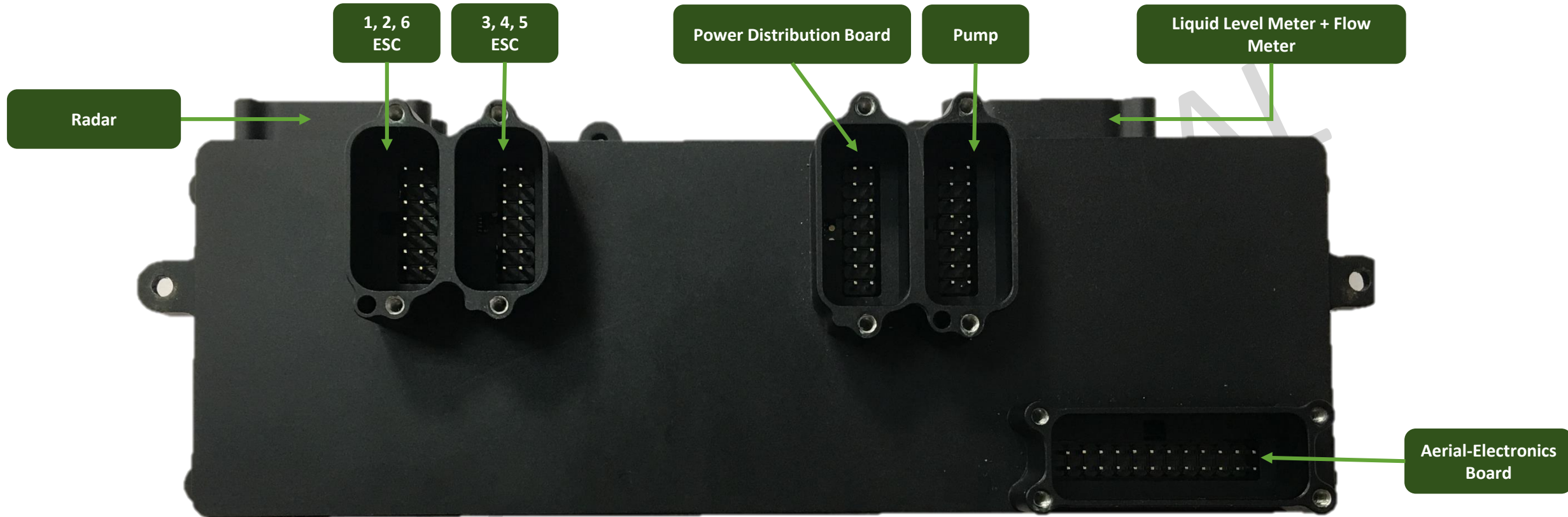
2. Aerial-Electronics Board and Spraying Board Port

3. Dual-Barometer



- 1. Update port:** Users can connect the PC to the aircraft via this port to update the firmware by DJI Assistant 2 for MG.
- 2. Aerial-Electronics Board and Spraying Board Port:** This port sends the spraying system's information, and it handles all modules that are connected to the spraying board, including the ESC, radar, flow meter, liquid level meter, and battery.
- 3. Dual-Barometer:** The Dual-Barometer receives the current barometric pressure signal, sends the barometric pressure information to the aerial-electronics board, and detects the aircraft's current altitude information.

Spraying Board - Back



The **Spraying Board** is the control system of the aircraft's core spraying function. It controls the pump and spraying volume. It also sends information from the ESC, radar, liquid level meter, and flow meter to the main controller.

Note: The compass board is inside of the spraying board.



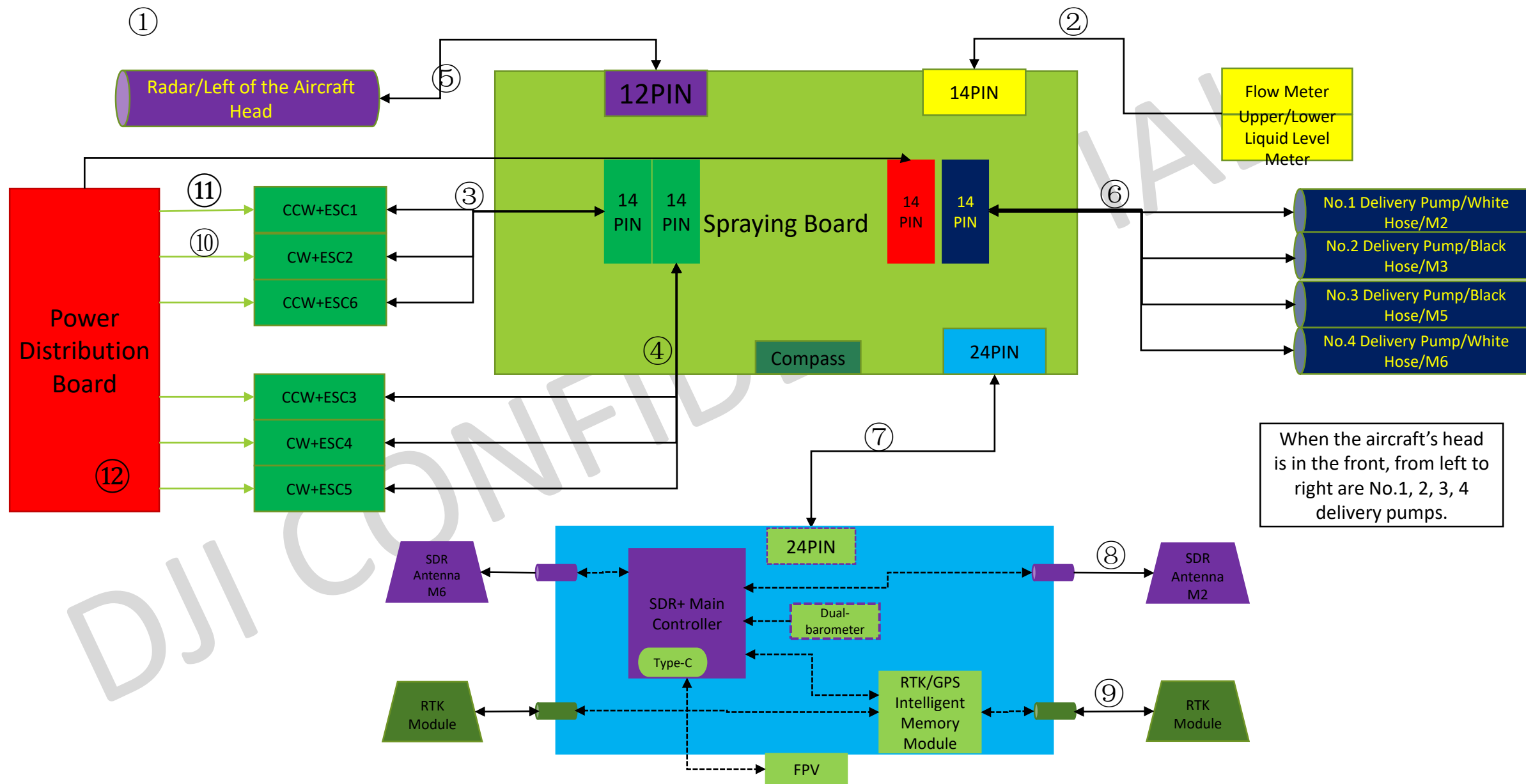
Module Principle

T16 Cable Connection

T16 Damage Assessment

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No.	Name	Function
1	Spraying Board_Power Signal Cable Connecting the Power Distribution Board and Spraying Board	Connecting the power distribution board with the spraying board.
2	Spraying Board_Liquid Level Meter Cable	The spraying board connects the flow meter and liquid meter.
3	Spraying Board_Signal Cable A Connecting the Spraying Board and Propulsion ESC	Connecting the No.1, 2, and 6 motors to the spraying board (there are raised dots on the port).
4	Spraying Board_Signal Cable B Connecting the Spraying Board and Propulsion ESC	Connecting the No.3, 4, and 5 motors to the spraying board (there are raised dots on the port.)
5	Radar_12 Pin Power Cable	Connecting the spraying board to the radar.
6	Spraying Board_Pump Cable	Connecting four pumps to the spraying board.
7	Aerial-electronics _Main Power Cable	Connecting the aerial-electronics board to the spraying board.
8	OcuSync_Antenna	The antenna is mounted on the arm, and it is linked to the aerial-electronics board. The M2's and M6's antenna are the same.
9	RTK Antenna Feeder	Connecting the aerial-electronics board to the RTK module. The left one and right one are the same.
10	Power Module_ESC Cable A	M6, M5, M3, M2
11	Power Module_ESC Cable B	M4, M1
12	T16 Power Distribution Board (Including ESC Power Cable)	Connecting the power distribution board to motor ESC.



Module Principle

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Damage Assessment - Visual Inspection Process

1. Check if the radar cable is damaged.

2. Check if the landing gear is damaged or misshapen.

3. Check if the battery is swollen or damaged.

4. Check if the spray tank leaks water.

5. Check if the T16 aircraft is damaged.

6. Check if the arm carbon tube is damaged or loose.

7. Check if the motors can rotate smoothly or if the surface is damaged.

8. Check if the propeller is damaged.



Check if the arm carbon tube and ESC cables are damaged.



Check if the airframe is misshapen or damaged.



The left picture shows a frame arm module: Check if the propeller is damaged, if the motor can rotate smoothly, if the ESC's surface is damaged, and if the arm carbon tube is cracked.



The right picture shows a spray lance: Check if the spray lance is cracked, if the nozzle is blocked up, and check if the hose is damaged.

There are six frame arms in total. M1 and M4 are short arms. M2, M3, M5, and M6 are long arms. The spray lance is mounted under the long frame arms as shown in the picture below.





Spraying Board

RTK Antenna

Front Frame

Middle Frame

Rear Frame

Power Distribution Board

After the frame arms are removed, the aircraft body and landing gears remain.

1. Check if the RTK cover and cables are damaged.

2. Check if the front, middle, and rear frames are damaged.

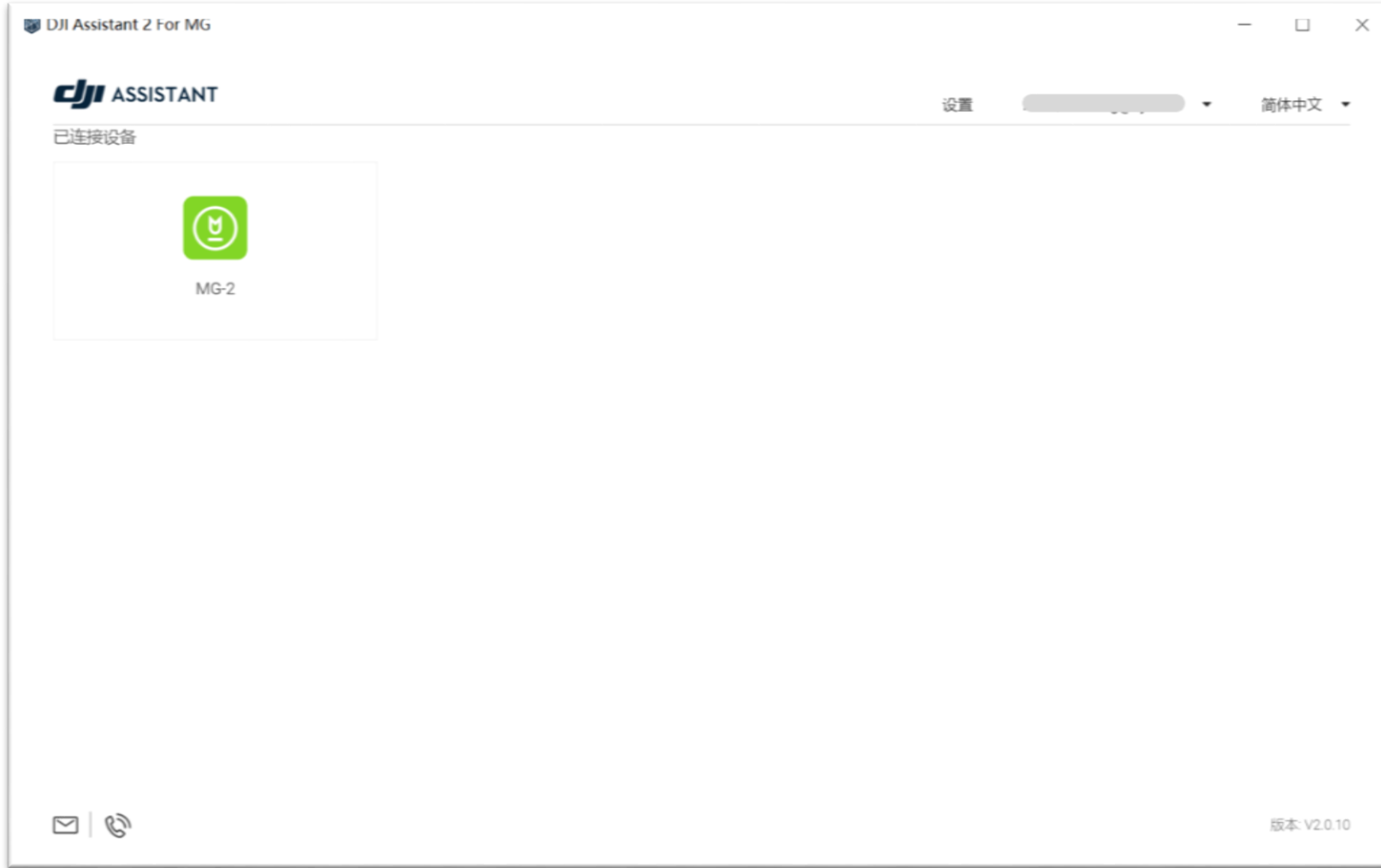
3. Check if the auxiliary structural components inside the aircraft body are misshapen or damaged.

4. Check if the cables are damaged.

5. Check if the radar is damaged.

6. Check if the landing gear is misshapen or cracked.

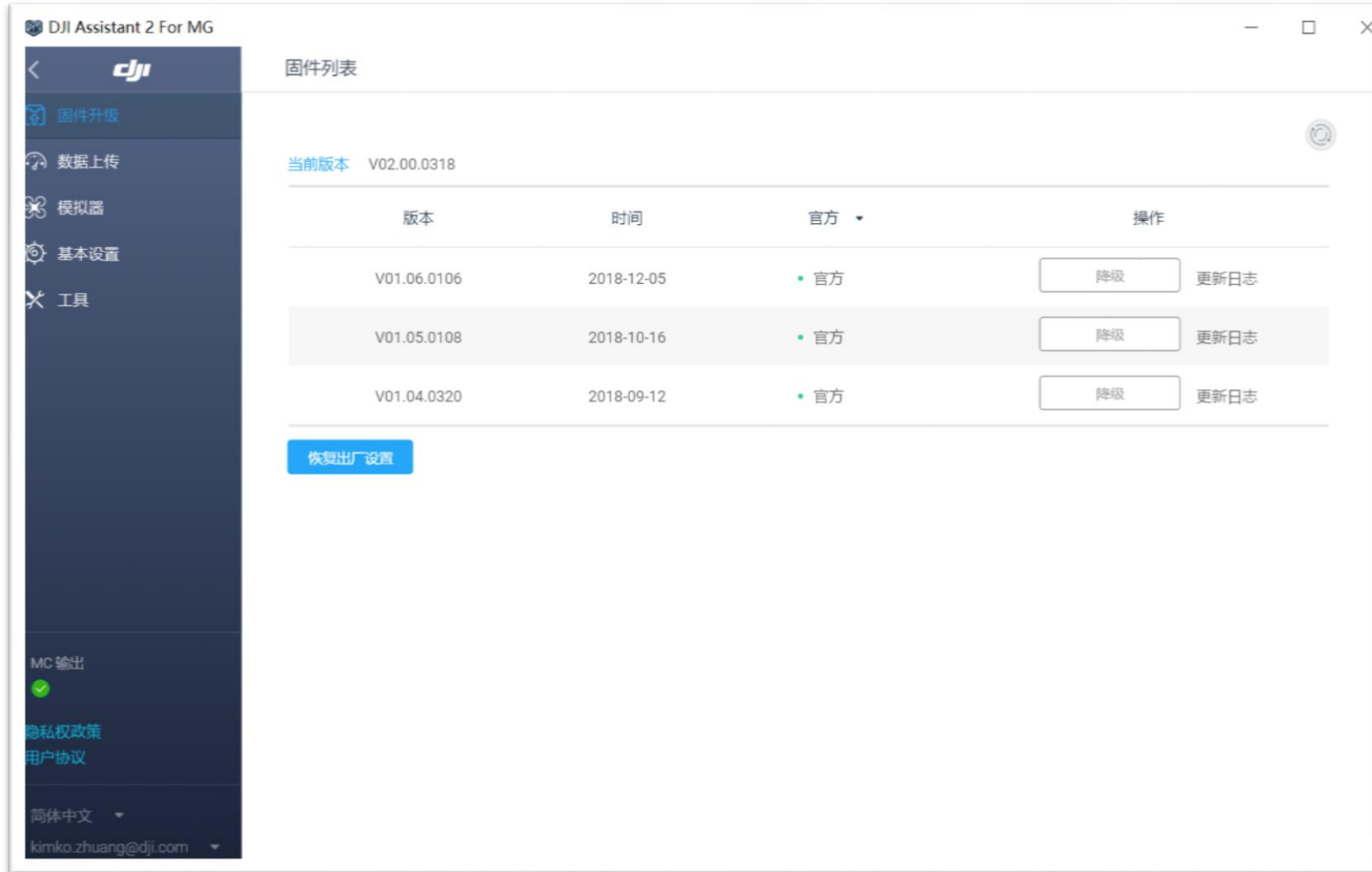
Damage Assessment Functionality Test - Tools (DJI Assistant 2 for MG)



Tools Needed: PC, Cable

1. If the aircraft does not have any cosmetic defect and can be powered on normally, the aircraft can be checked by DJI Assistant 2 for MG.
2. Connect the aircraft's USB-C port (under the waterproof cover) to the PC, launch DJI Assistant 2 for MG, and click "MG-2" to enter the main interface.

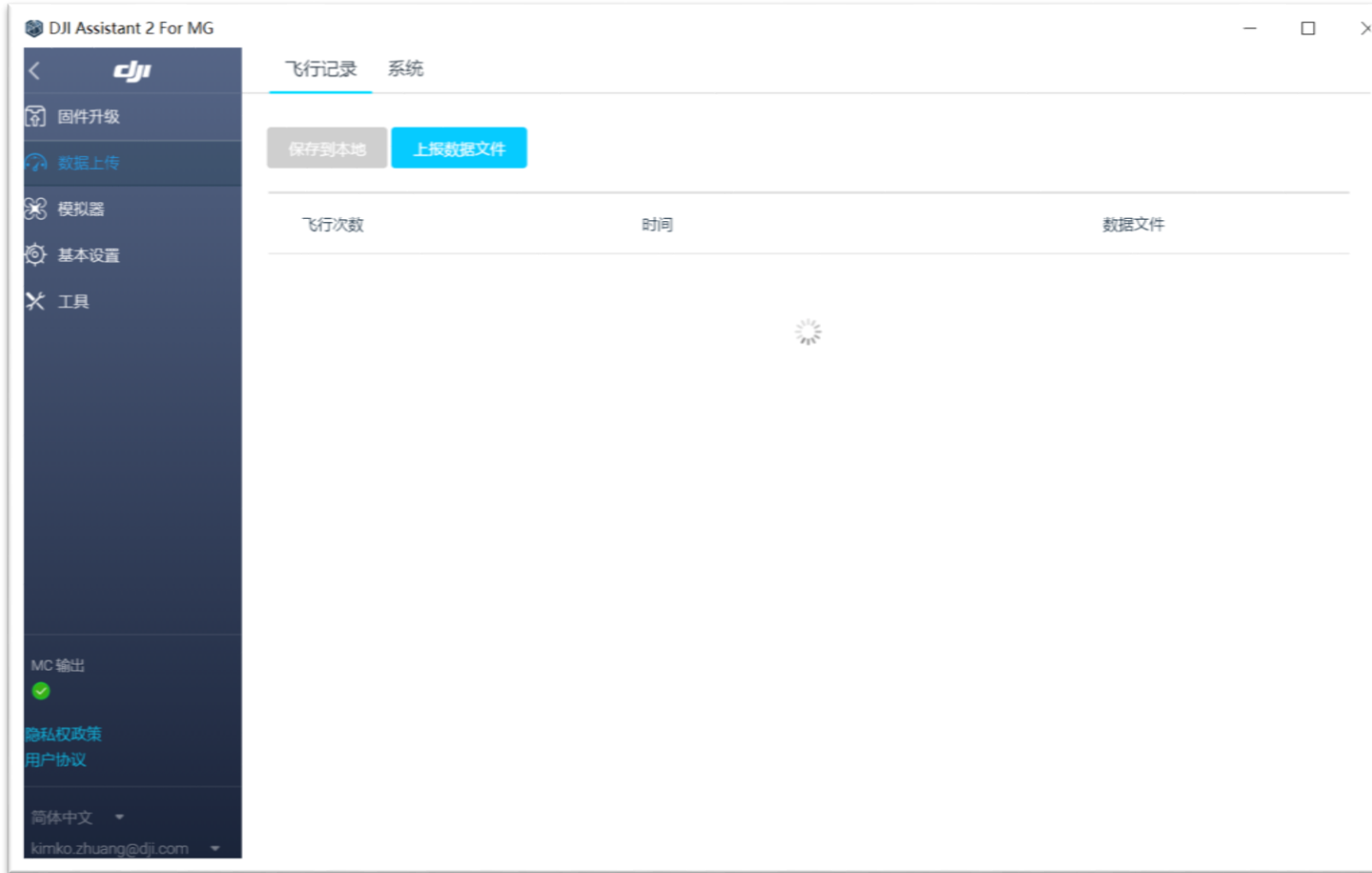
Functionality Test - DJI Assistant 2 for MG 2.1



Firmware Update -> Check if the aircraft needs to be updated in the firmware update list.

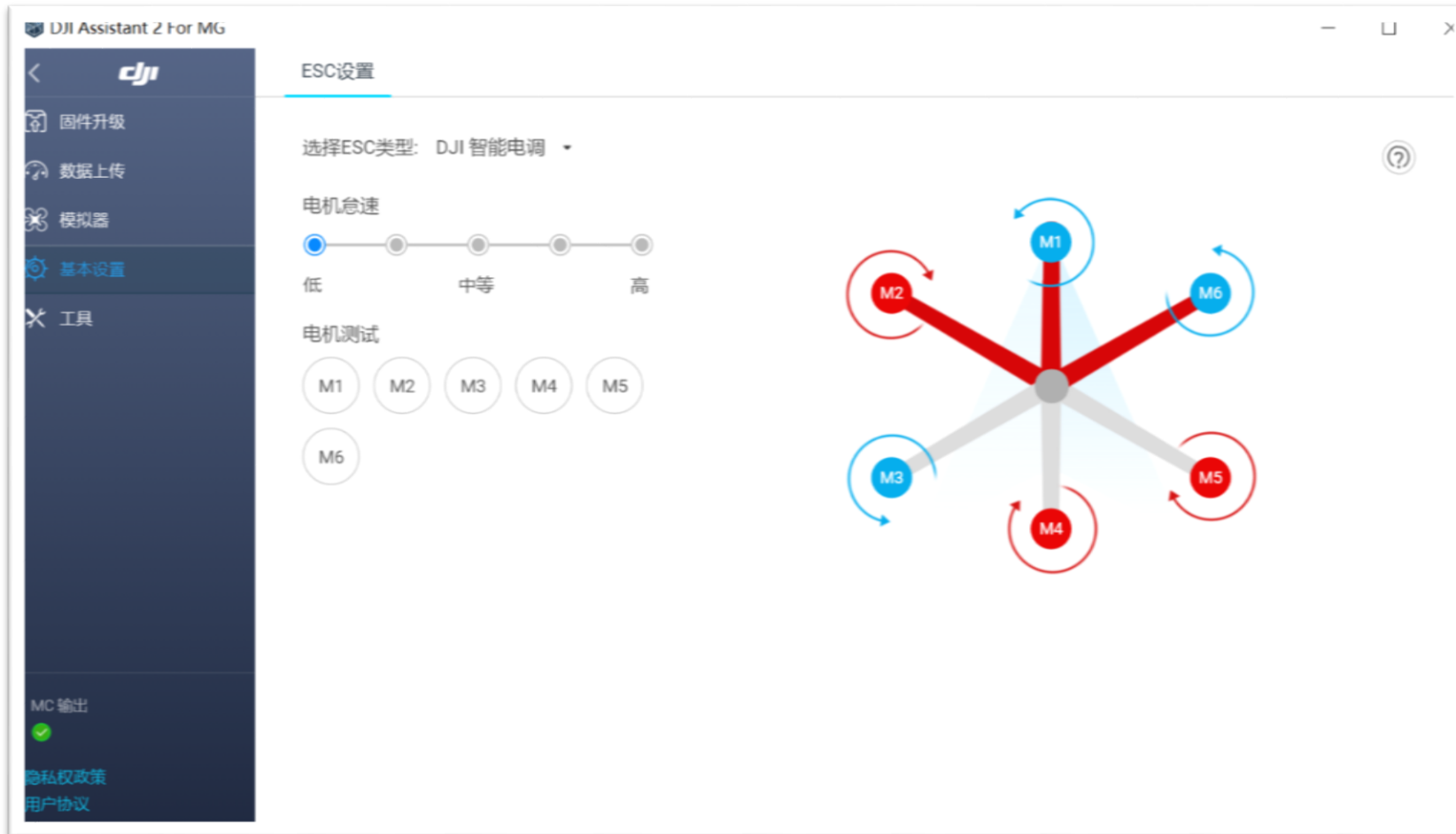
Users can check the updated content of the new firmware in the update log.

Functionality Test - DJI Assistant 2 for MG 2.2



Log Export: Users can check the aircraft's flight record, and they can export the flight record for analysis.

Functionality Test - DJI Assistant 2 for MG 2.3

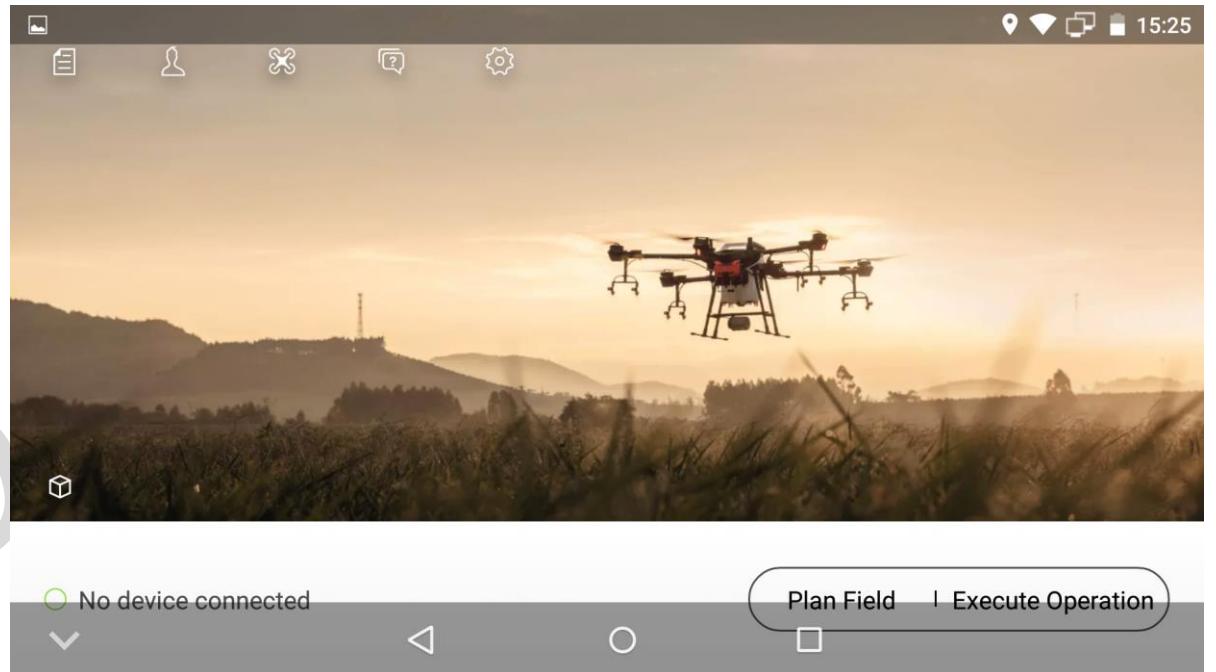
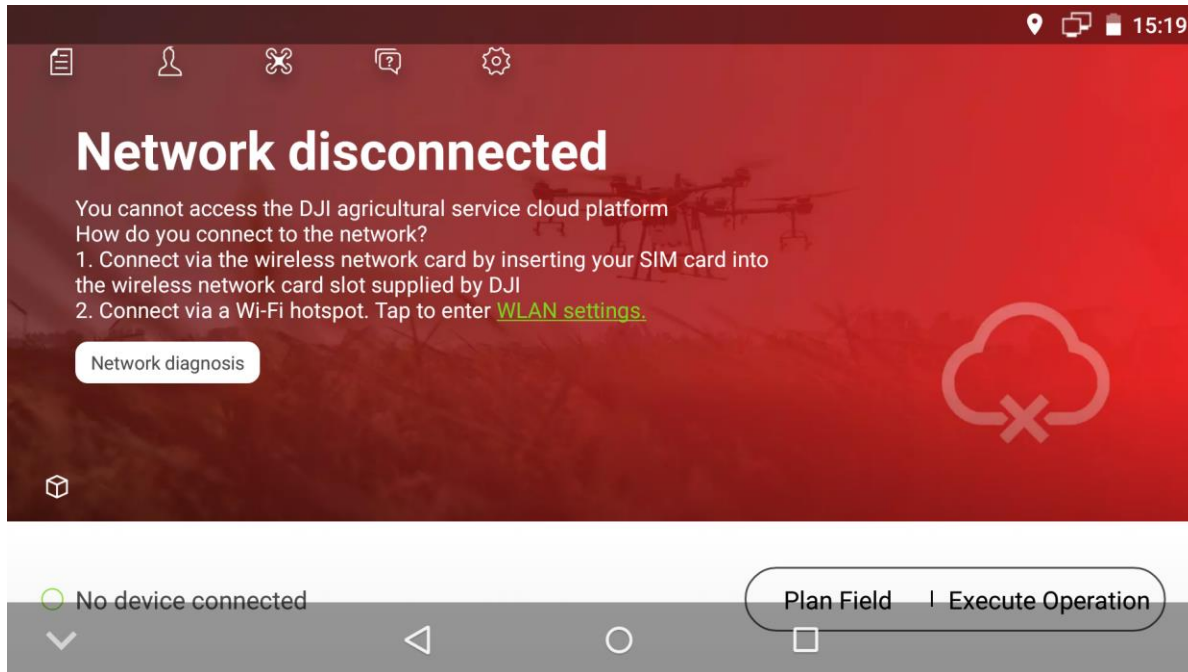


Basic Settings ->ESC Settings
Check the M1-M6 motors' rotation.

M1/3/6 motors rotate anticlockwise.

M2/4/5 motors rotate clockwise.

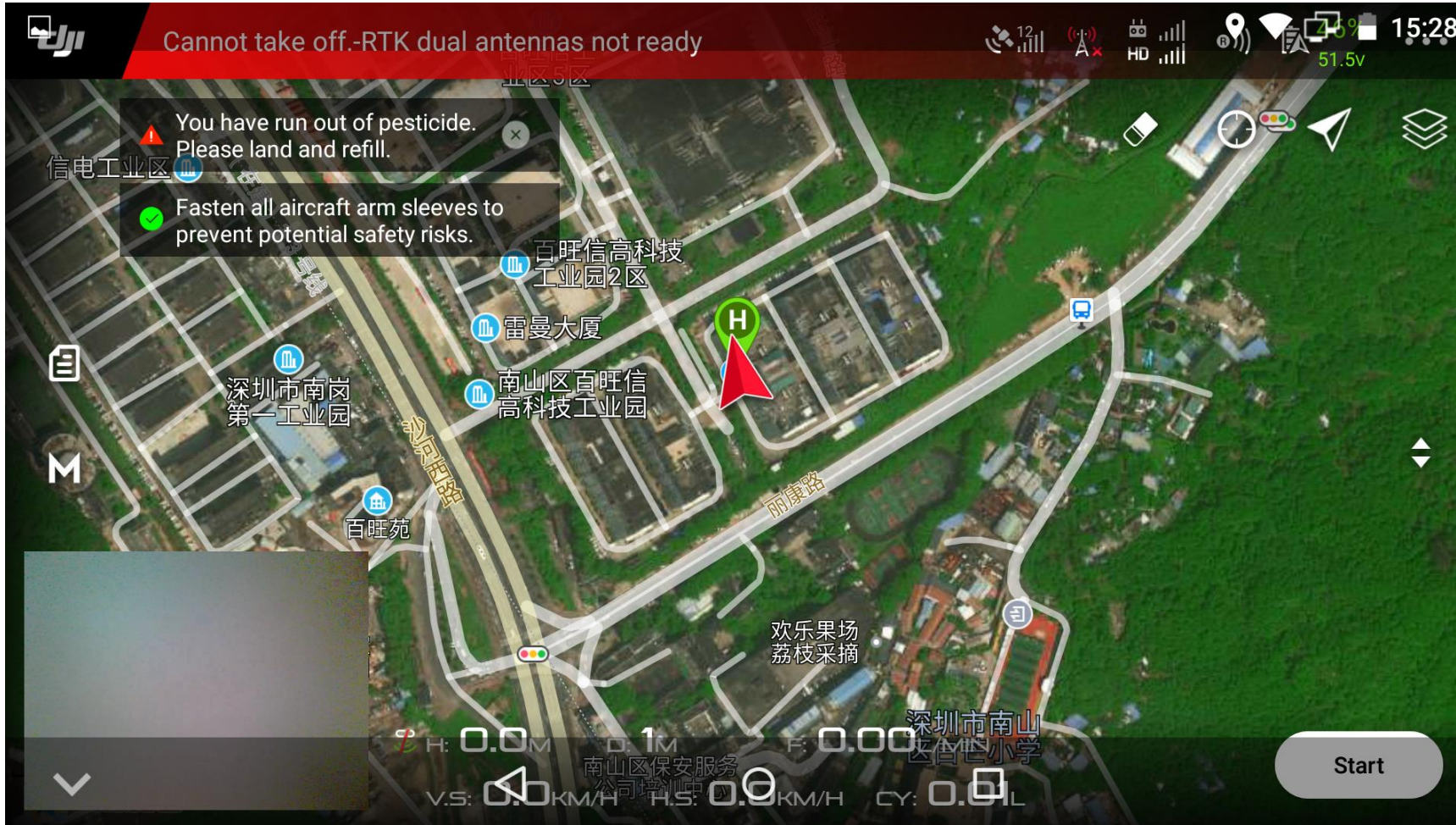
Functionality Test - Tools (Remote Controller)



The aircraft's function errors cannot be recognized by the naked eye. In this case, tools (like a remote controller) can be helpful. Link the remote controller to the aircraft, then perform the check according to the remote controller's error reports.

*The T16's and MG-1P's remote controller are the same, but the firmware is different.

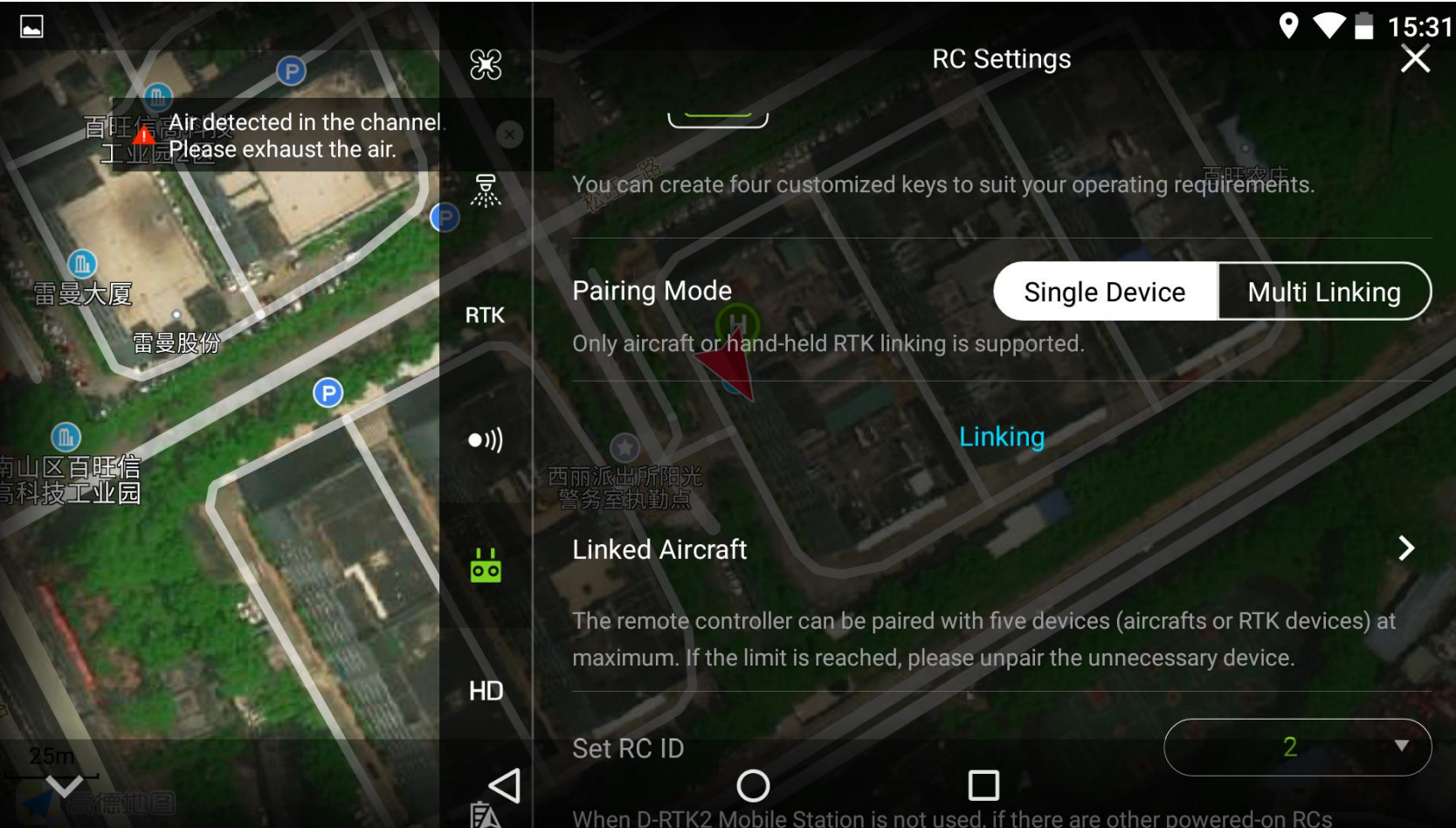
Functionality Test - Tools (App)



The remote controller can offer a lot of information, such as aircraft model, aircraft version, owner, operation MU, battery level information, control stick mode, if the spray tank has been installed properly, and so on. The remote controller is essential to damage assessment.

The picture above shows errors. Therefore, we can locate the related modules

Functionality Test - Tools (App)

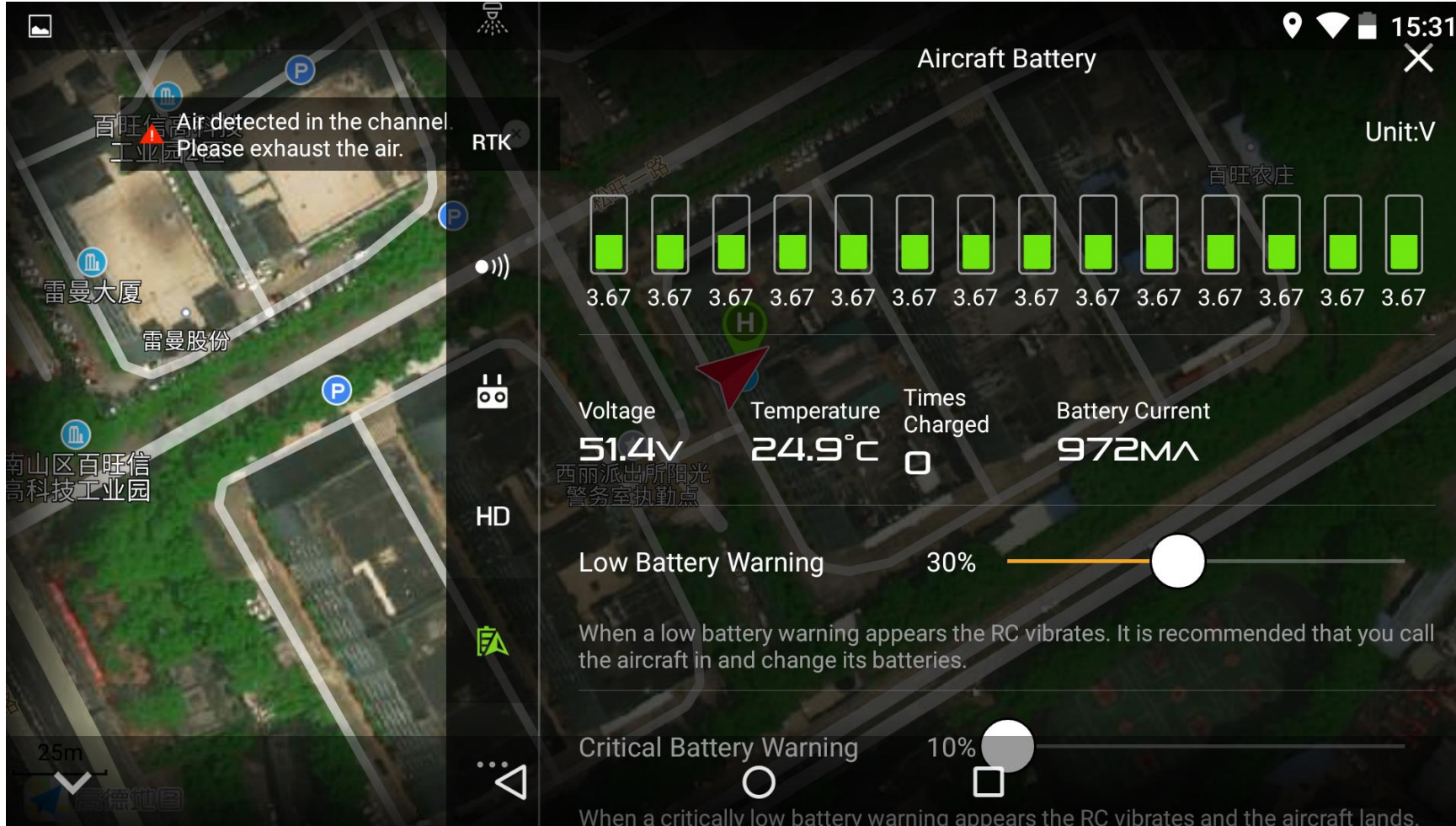


The left picture shows what happens after the remote controller's app triggers linking.



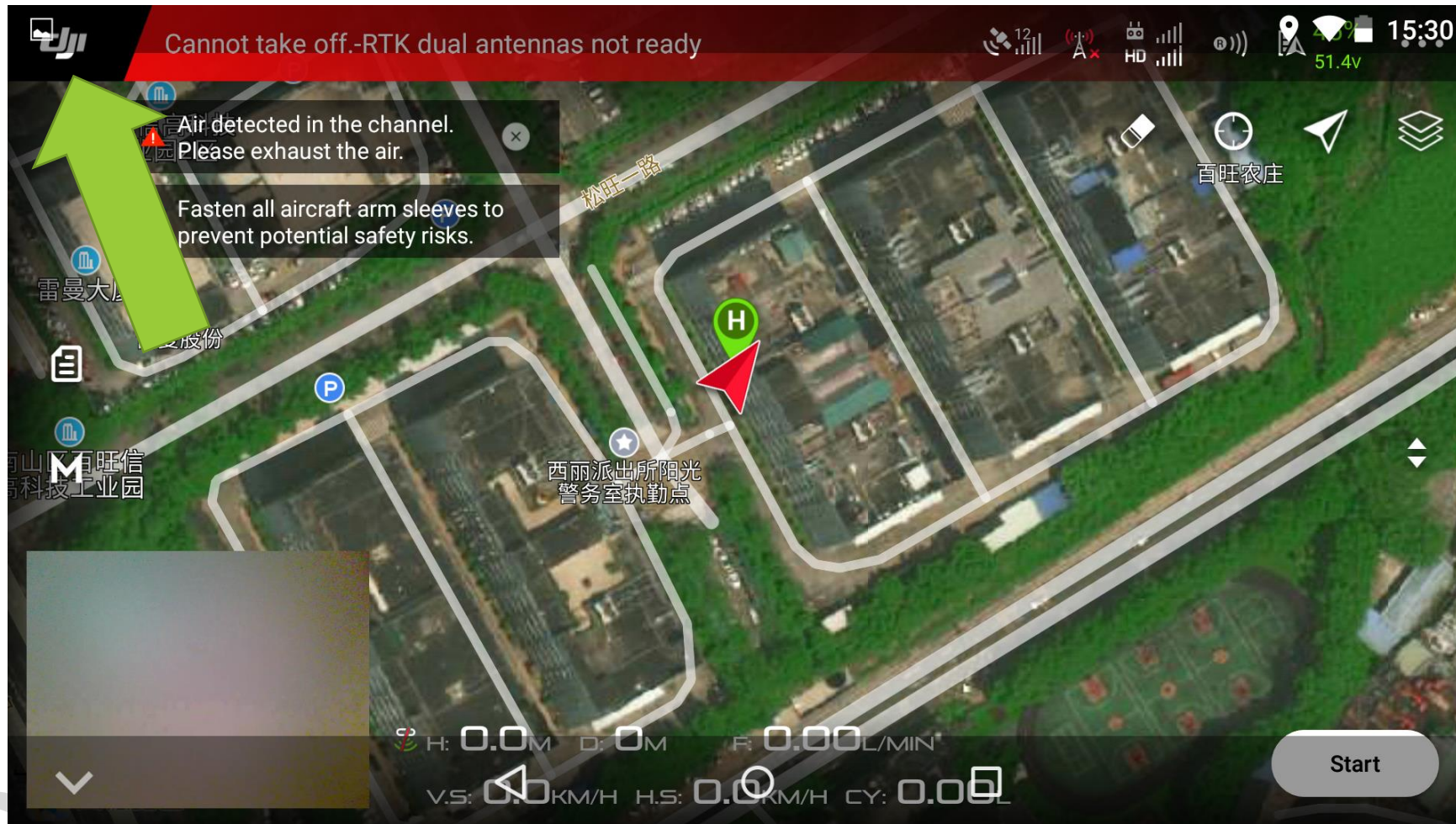
Right picture: Press and hold the aircraft's power button for five seconds, and then the linking starts.

Functionality Test - Tools (App)



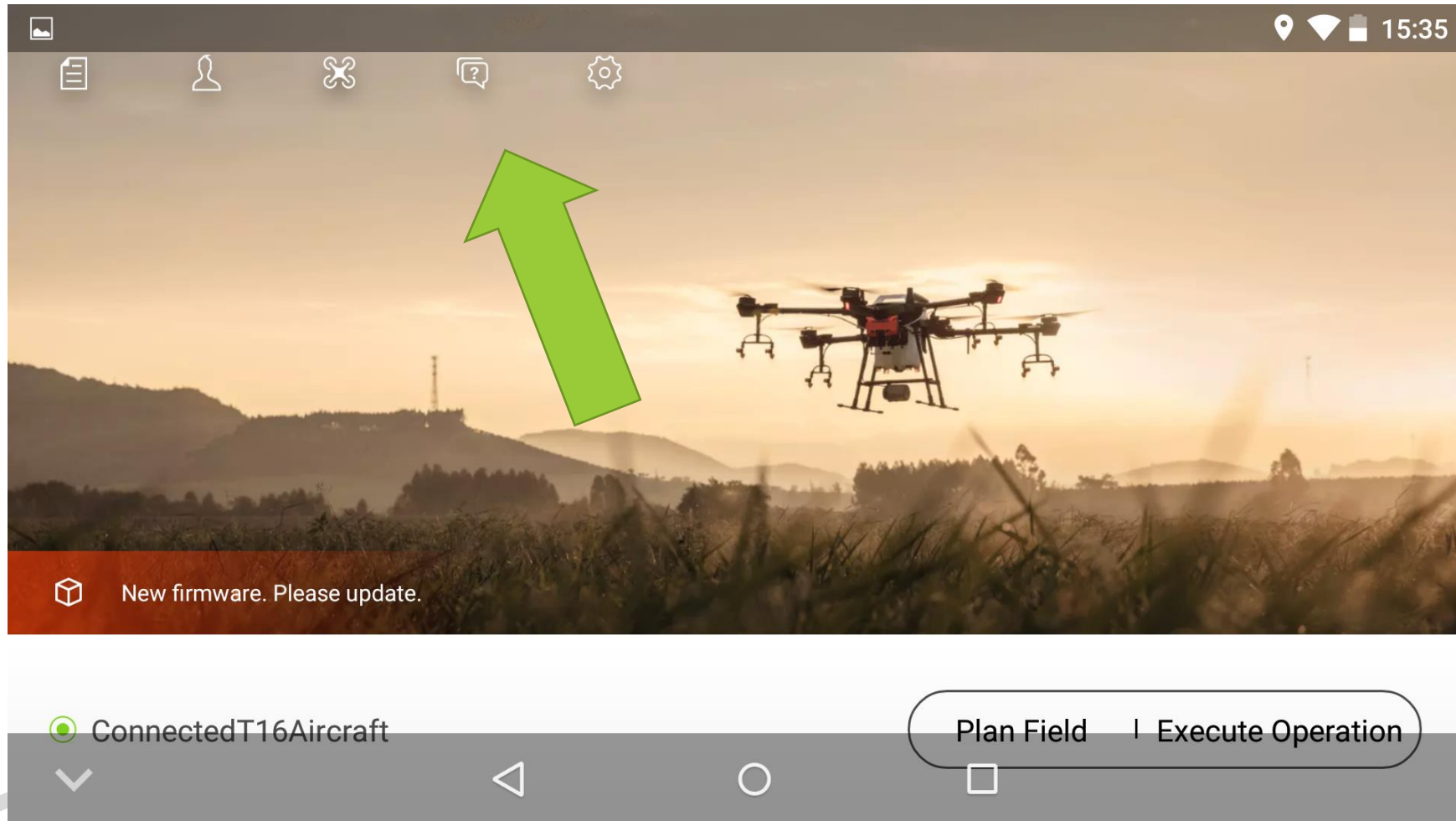
From the app you can learn information such as there are the amount of battery cells (14), battery cycle times, battery levels, and so on.

Functionality Test - Linking Failure 1



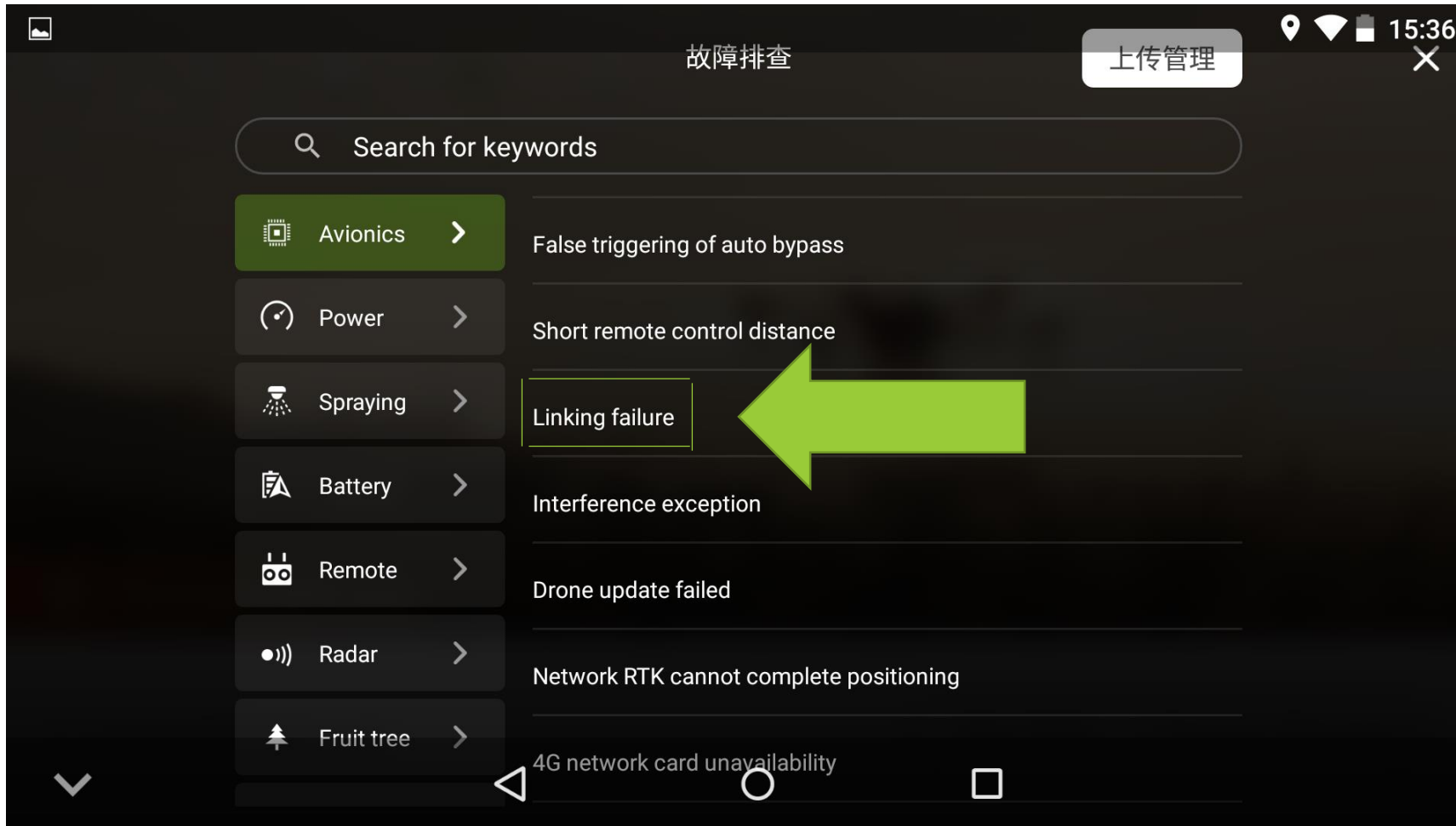
Click "DJI" and back to the lobby

Functionality Test - Linking Failure 1



Enter error logging menu

Functionality Test - Linking Failure 1



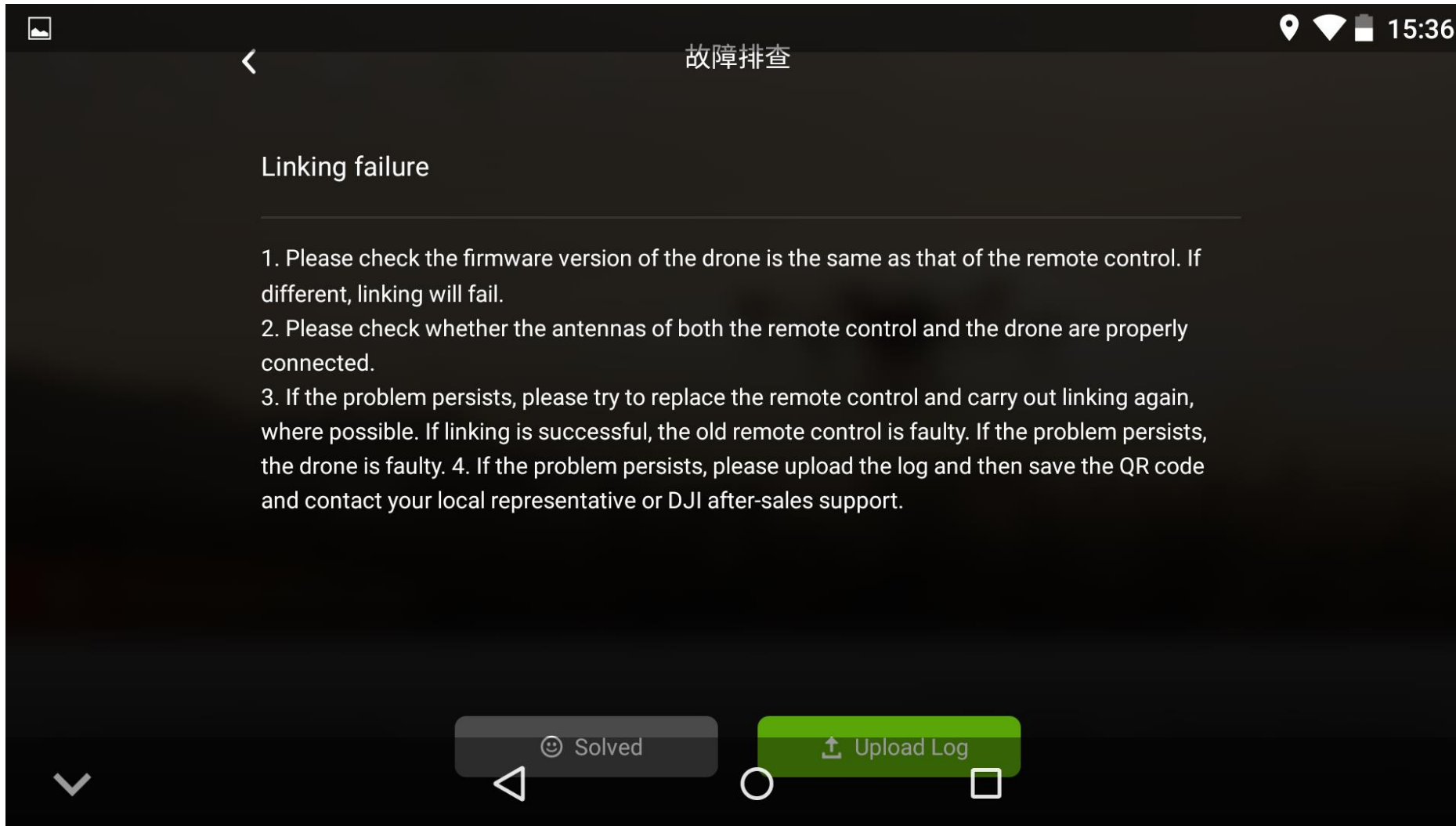
If users' aircraft cannot link to the remote controller:

Troubleshooting Process

1. Check if the remote controller's and aircraft's versions are consistent.
2. Fix the issue according to the app's error report. (Take screenshots or record the operation process.)
3. If the issue still cannot be fixed, contact customer service at support.agras@dji.com and upload the log.

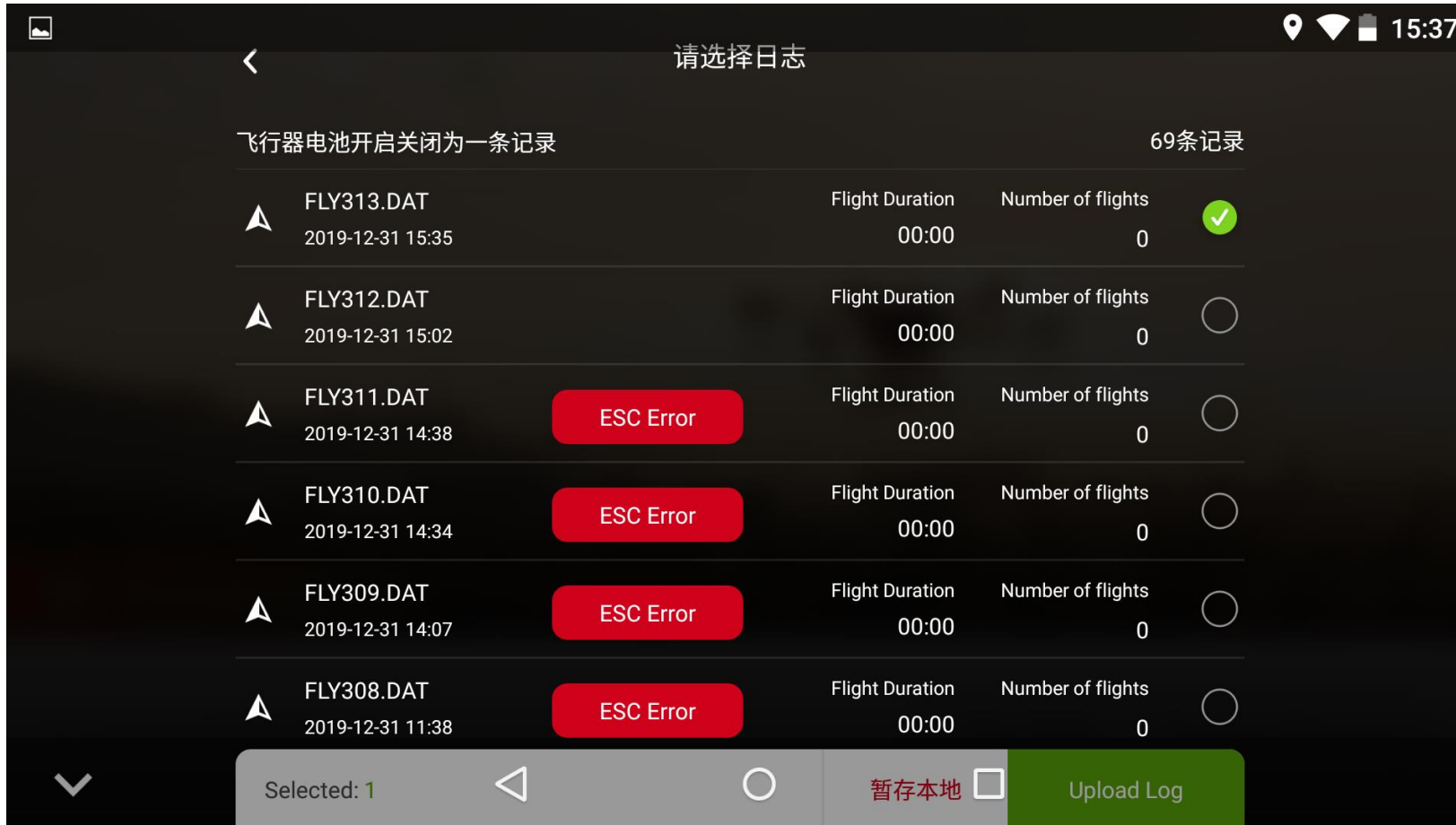
As shown in the picture, check the error reports framed in the **No.1 box**, and then fix the issue according to the prompt. If the issue cannot be fixed, then send the log as shown in the **No.2 box** to support.agras@dji.com.

Functionality Test - Linking Failure 2



Users can troubleshoot and fix the issue based on the error report.

Functionality Test - Linking Failure 3



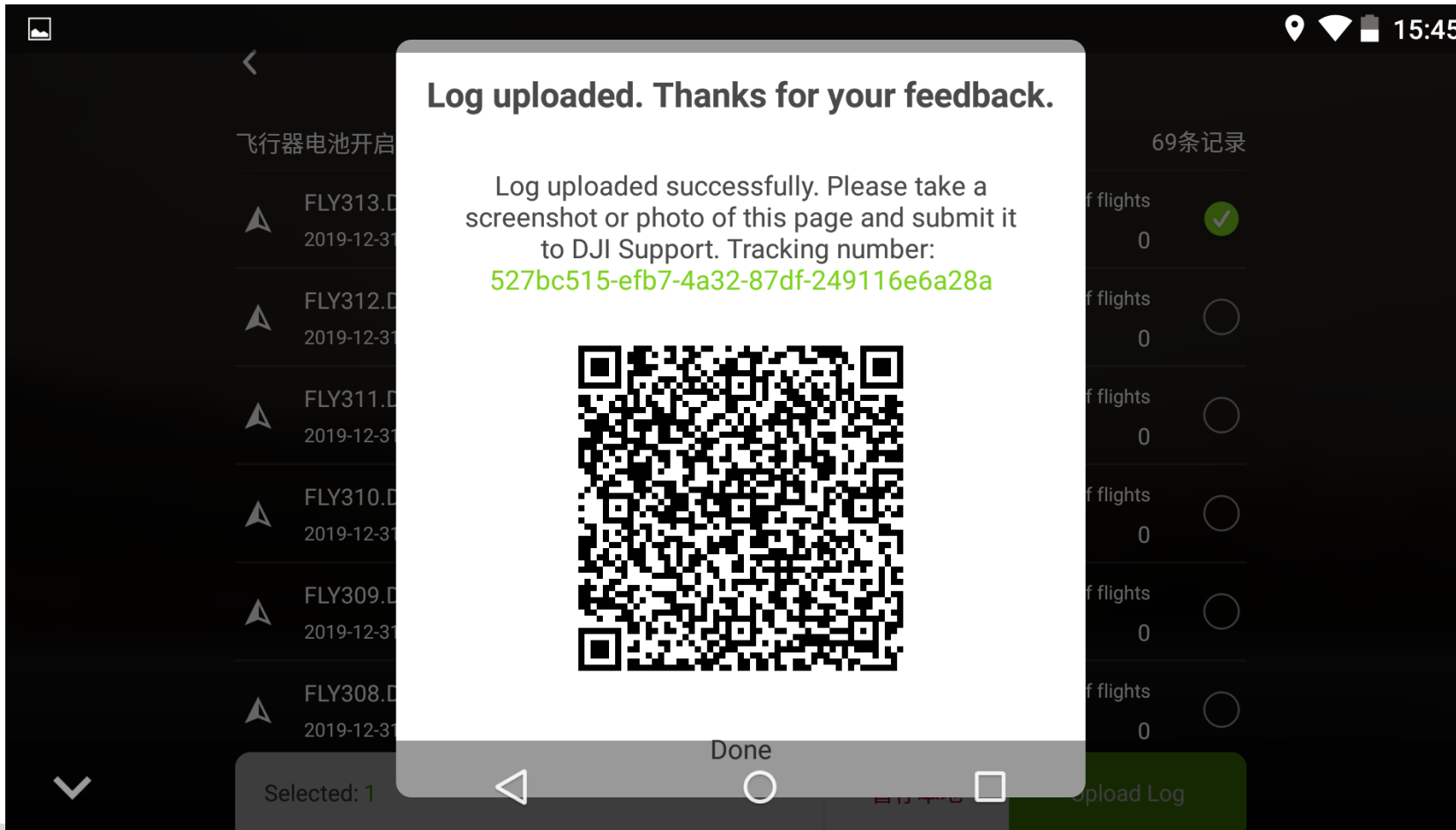
Choose the error report type, upload the data, and then report the feedback to support.agras@dji.com.

If there are multiple logs, then users need to choose the log manually according to time before uploading it.



- app must be connected to internet.

Functionality Test - Obtaining Data QR Code



Note: Save the QR screen shot for feedbacks



Module Principle

T16 Cable Connection

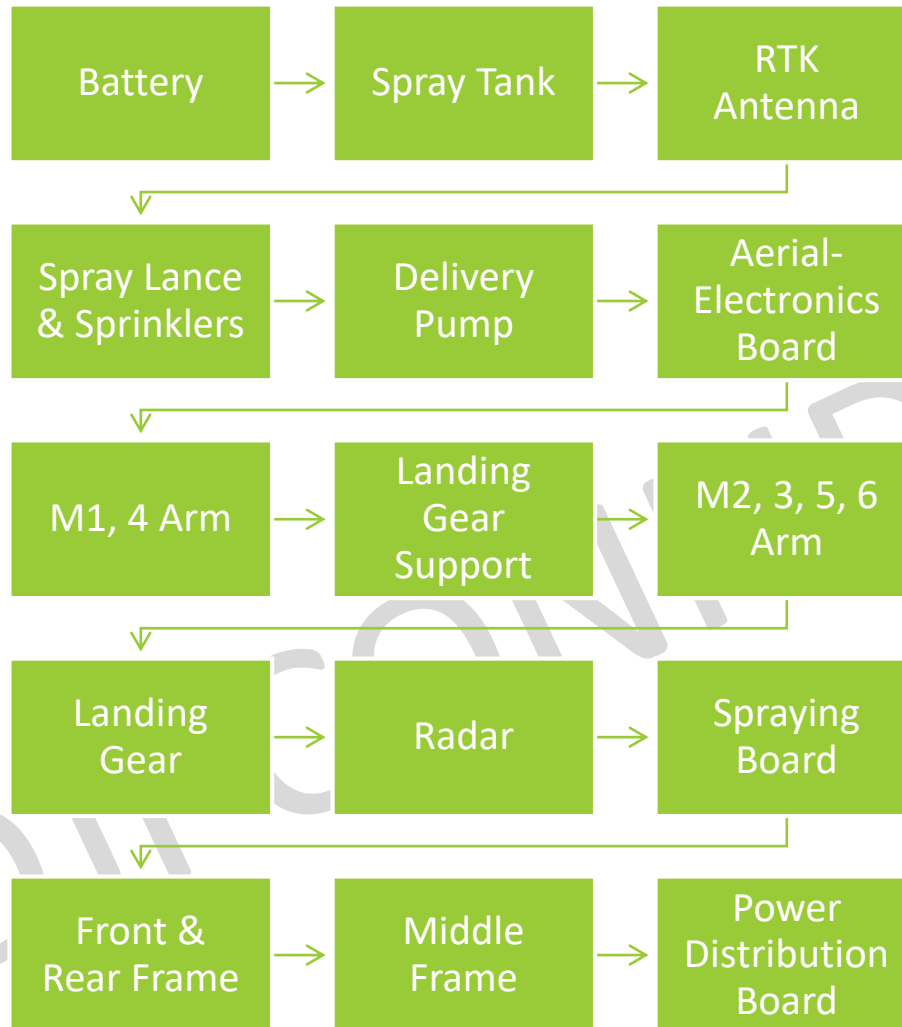
T16 Damage Assessment

T16 Disassembly

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Repairing Whole Device

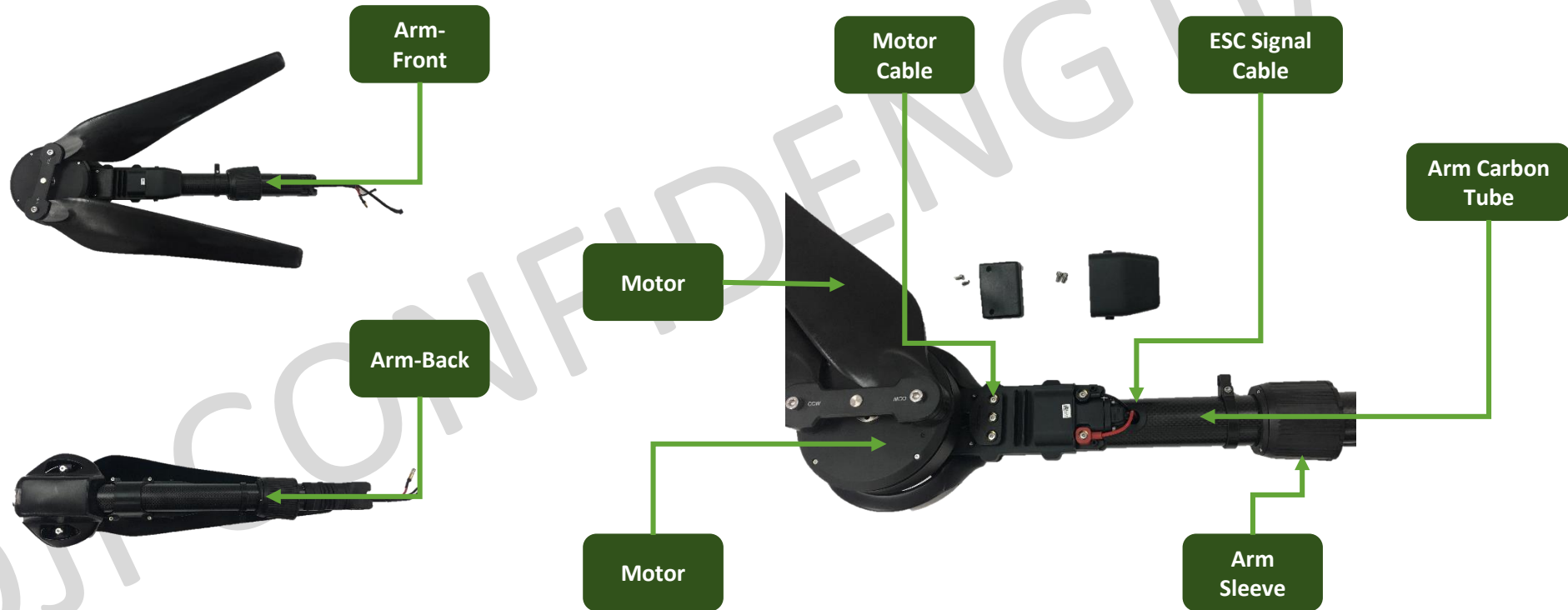
Tools and Assembly & Disassembly Process



Tools for Assembly/Disassembly	Specification
4 mm Hexagonal Wrench	H4.0 Hexagonal Wrench
Star Socket Screw Bit	T6
Hexalobular Socket Screw Bit H2	H2
Hexalobular Socket Screw Bit H2.5	H2.5
Hexalobular Socket Screw Bit H3	H3
Hex Sleeve	M10
Hexalobular Socket Screw Bit H1.5	H1.5

Repairing Parts - Aircraft Arm

The M1 and M4 arms are short, the M2, 3, 5, and 6 are long arms. The six ESCs and six motors are interchangeable.



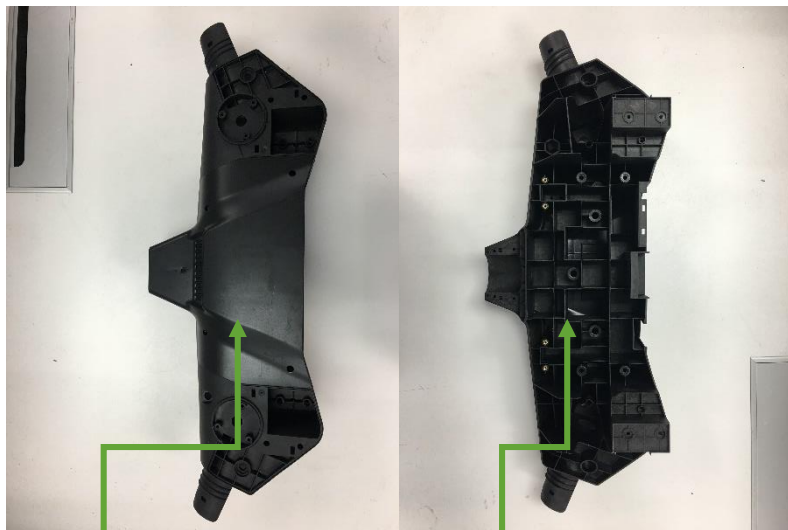
Repairing Parts - Aircraft Body

The aircraft body is composed of a front, middle, and rear frame.

The front frame is located at the front of the aircraft, and it is connected to the M1 arm.

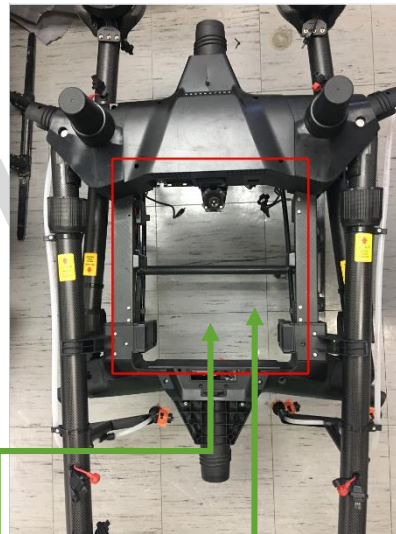
The middle frame is formed by two identical L-shaped carbon tubes.

The rear frame is located at the rear of the aircraft, and the power distribution board is mounted inside the frame.



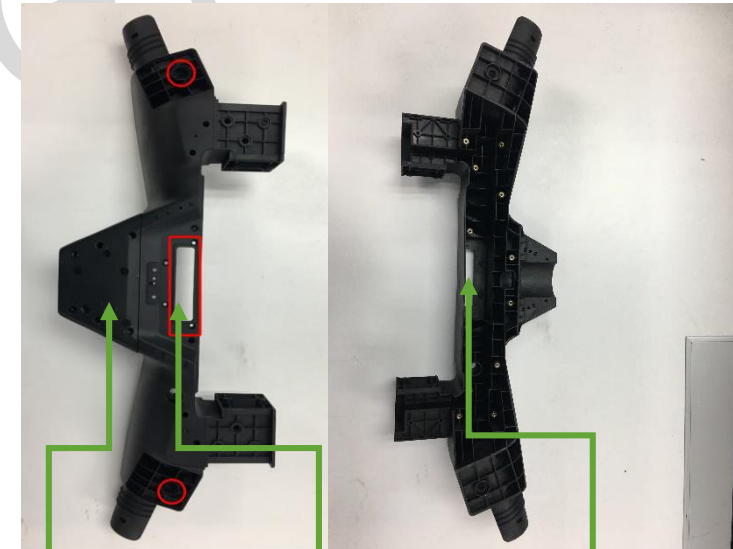
Mount the
spraying
board.

Front Frame



Place the spray
tank and
battery in the
frame.

Middle
Frame



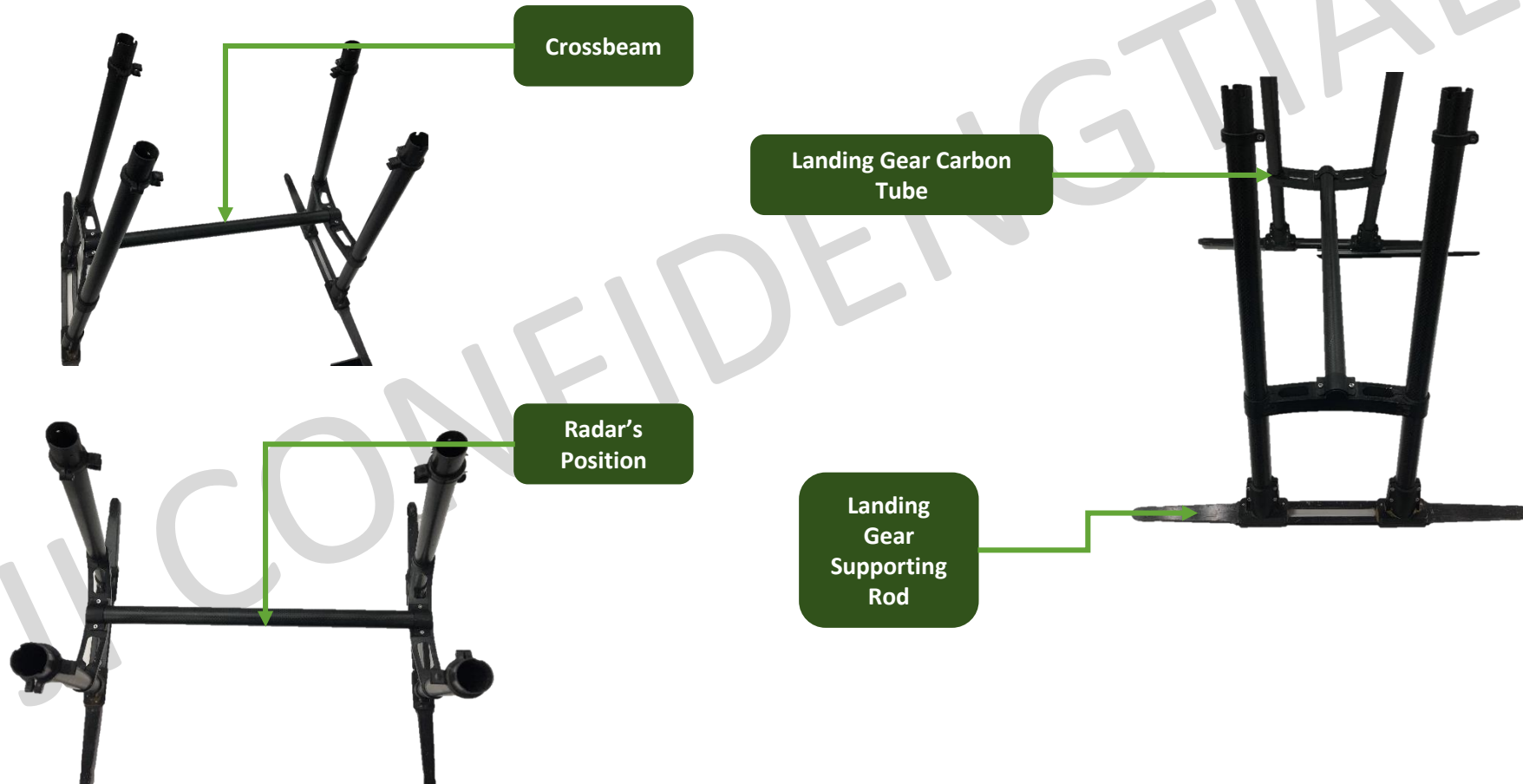
Rear Frame

Mount the
power
distribution
board.

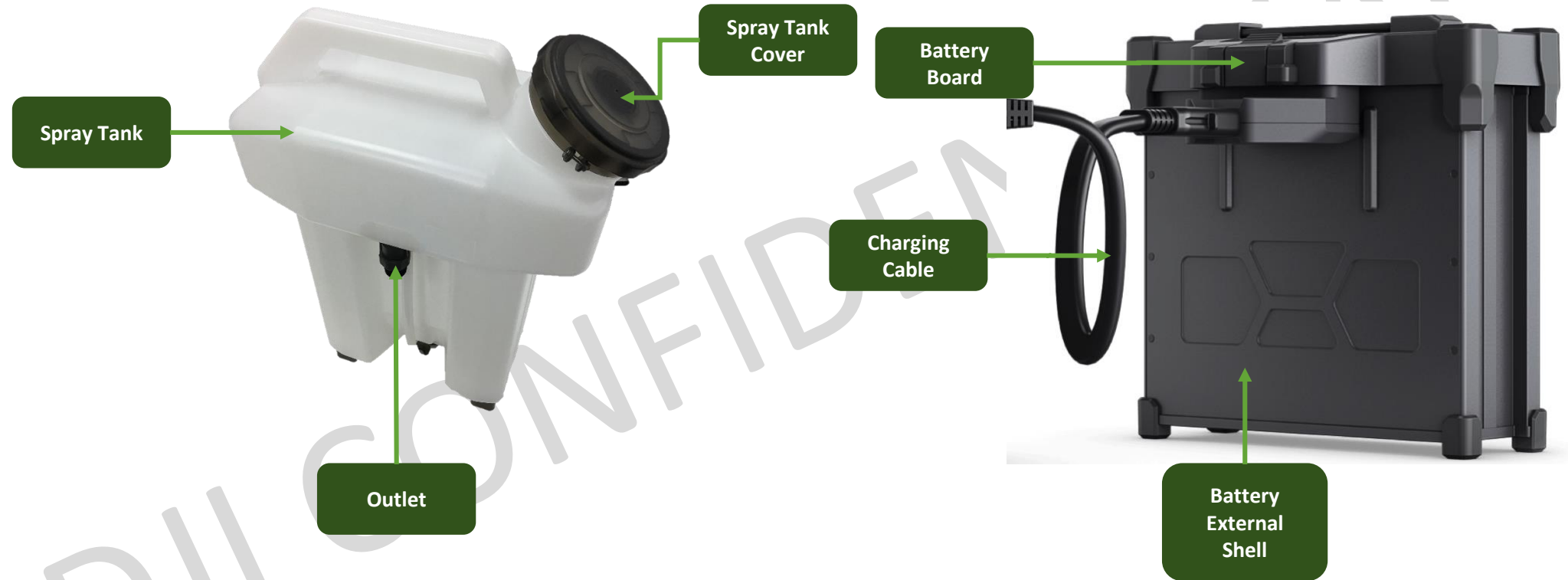
Mount the power
distribution board.

Repairing Parts - Landing Gear

The left and right landing gears are the same. The two landing gears are connected by a crossbeam, and a radar is mounted on the crossbeam.



Repairing Parts - Spray Tank and Battery





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T16 Aircraft Status Indicator Description



The aircraft status indicators are the No.3, 4, and 5 indicators. They are at the rear of the aircraft.

- After the aircraft reaches its maximum altitude, the status indicators will blink red and green slowly, and the three front LEDs will glow red.
- If the remote controller is not linked to the aircraft, the aircraft status indicators will blink yellow rapidly.
- After the aircraft enters the linking state, the status indicators will blink yellow rapidly, and the three front LEDs will glow solid red.
- When the flying condition is normal, the status indicators will blink green, and the three front LEDs will glow solid red.
- When the IMU is being calibrated, the status indicators will blink green rapidly. (Only the polyhedron version is available. The monohedron version has been canceled.)
- When the battery level is low, the status indicators will flash red to report an alarm. When the battery level is extremely low, the indicators will flash red rapidly.
- When the aircraft is self-checking after it is powered on, the status indicators will blink red, green, and yellow alternatively.
- Point A recorded: The indicators will blink red four times.
- Point B recorded: The indicators will blink green four times.
- Solid green: A serious error has occurred.

1. What happens if the T16's 2600W 4-channel intelligent battery charger's charging indicator blinks yellow?

Blinks yellow three times continuously: The battery's temperature is below 0°C, and the battery charger has stopped charging. (If the battery is being charged when its temperature is lower than 0°C, the battery will swell up and be damaged.)

Blinks yellow once continuously: The battery's temperature is low (0-5°C), and the charging current is limited.

Blinks yellow four times continuously: The battery's charging temperature is too high, and the battery charger has stopped charging. (This issue may occur if the battery is charged instantly after flying.)

2. What is the T16's 2600W 4-channel intelligent battery charger's charging logic? Is the battery cell voltage difference balancing improved?

The battery will not balance the battery cell voltage difference. The batteries themselves balance this difference.

3. Can the 2600W 4-channel intelligent battery charger charge different batteries?

The 2600W 4-channel intelligent battery charger cannot charge different batteries. The batteries being charged should all be first-generation batteries or second-generation batteries. For example, if MG-1P batteries and T16 batteries are being charged together, the charger will indicate that a communication error has occurred.

4. What is the T16's 2600W 4-channel intelligent battery charger's charging logic?

When the charger is being powered on, the default charging mode is slow charge mode. Users can press the Mode button to switch between quick charge and slow charge mode.

Charging logic: In slow charge mode, if the four batteries' battery levels are different, then the charger will charge the battery with the lowest battery level first. After this battery's battery level becomes the same as the battery with the second lowest level, the two batteries will be charged together.

When the charger is in the quick charge mode, the batteries will be charged one by one, starting from the one with the highest battery level.

When the charger is in the slow charge mode, the batteries will be charged together after the battery with the lowest battery level has been charged.

If a new battery is inserted into the charger when the charger is in quick charge mode, then the charging condition will not be affected. If the charger is slow charge mode and a battery is inserted or removed, the charger will choose a new battery to charge.

5. Can the T16's battery be charged by a previous product's charger?

No, the T16's battery cannot be charged by a previous product's charger. But the T16's battery charger can charge MG-1P/MG-1S's batteries.

6. How long does it take to fully charge the T16 in different charging modes?

Quick charge mode: 20 minutes. In this mode, the batteries are charged one by one, and the battery with the highest battery level will be charged first.

Slow charge mode: It takes 66 minutes to fully charge one battery if it is charged separately. It takes 80 minutes to fully charge four batteries if they are charged simultaneously.

1. Can the T16 aircraft still operate if the flow meter and liquid level meter are damaged?

If the app reports an error by mistake, the aircraft can operate normally.

Users can cover the sensor to make the sensor stop working so that there will be no error reports.

The aircraft can still fly if no spray tank is attached.

2. T16 Protection Rating

Radar/Aerial-electric System/Spray Control System/Propulsion ESC: IP67 Aircraft: IP54

The battery connector, motor front frame upper cover, and RTK antenna cannot be rinsed by water.

3. After the remote controller is powered on, it reports an alarm and the indicator flashes red.

If the remote controller's control stick is not in the center place, calibrate the control sticks after the remote controller is powered on.

If the battery level is below 10%, the remote controller will signal an alarm.

Before the aircraft is taking off, if the aircraft's temperature is above 70°C, the remote controller will signal an alarm. If the aircraft's temperature is above 80°C when it is flying, the remote controller will signal an alarm.

4. Under what circumstances will only four of the T16's sprinklers work? Under what circumstances will all of the sprinklers work?

When the flow is not higher than 1.8L/min and two pumps are working, only four sprinklers will work.

When the flow is higher than 1.8L/min, all sprinklers will work.

5. How does a user operate the T16 aircraft when using the D-RTK 2 handheld mobile station to record location information?

When using the D-RTK 2 handheld mobile station to record location information, the remote controller's app should be used. There are two operation modes based on the RTK mobile station's service:

A. When using the network RTK or the NTRIP service, the remote controller and D-RTK 2 mobile station are needed. Users need to switch the D-RTK 2 mobile station to handheld mapping device mode, and then enter the mapping page to record the location.

B. When using the D-RTK 2 mobile stations as a reference station and mobile station, two D-RTK 2 mobile stations and one remote controller are required. One D-RTK 2 mobile station should be switched to mobile base station mode, and the other D-RTK 2 mobile station should be switched to handheld mapping device mode. After the linking is finished on the remote controller, perform the operation on the app.

- 6. If the battery will not be used for more than ten days, please discharge the battery's battery level to 50%-65%, and then the battery's life time can be extended. The battery has saving and protection functions.**

If the battery's battery level is above 65%, then the battery will discharge automatically, and it can be stored naturally.

If the battery's battery level is below 50%, then users should charge the battery level to 50%-65% before storing the battery.



**THANK
YOU**